

Tidewater Port Authority Compliance Assessment: Stormwater Discharge Permit Compliance at Berth 14

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Executive Summary

This compliance assessment evaluates the Tidewater Port Authority's compliance with the industrial stormwater discharge permit that covers the Berth 14 Container Terminal for the permit year from 1 March 2025 to 28 February 2026. The assessment is required annually by the permit and is submitted to the state environmental agency.

The assessment found that discharges from the terminal exceeded the zinc limit in a quarter of the monthly samples, that outfall telemetry was unavailable for extended periods, and that some spill response kits were incomplete. Laboratory reporting met the permit deadlines throughout the year.

Four findings and three recommendations are presented. The Director of Terminal Operations has accepted all three recommendations. The overall assessment rating is that the terminal is not in full compliance and a corrective action plan is required.

Background

The Berth 14 Container Terminal is a 62-acre facility that handles about 540,000 containers a year and drains to the harbour through six permitted outfalls. Runoff from the container stacking area carries zinc from galvanised container surfaces and tyre wear, which is the pollutant of greatest concern under the permit.

Permit compliance is managed through the Stormwater Management Program, the authority-wide program that maintains the pollution prevention plan, schedules sampling and trains terminal staff. Flow and turbidity at each outfall are measured by the Harborwatch Monitoring System, a telemetry system that reports readings every fifteen minutes to the environmental team.

Laboratory analysis of the monthly samples is performed by Coastal Environmental Testing, an accredited laboratory vendor under a three-year contract. The permit requires results to be reported within 30 days of sampling.

The assessment reviewed all monthly sample results, the telemetry availability logs, the spill kit inspection records and the training register, and walked the terminal drainage system with the terminal environmental coordinator.

Findings

Finding 1: Stormwater discharge samples at the Berth 14 Container Terminal exceeded the permit limit for zinc in three of twelve monthly samples. Severity: High. The exceedances occurred in October, November and January, all months with heavy rainfall, and all at the outfall that drains the container stacking area. The permit requires the authority to implement additional treatment when a limit is exceeded in more than two samples in a permit year.

Finding 2: The Harborwatch Monitoring System lost telemetry from two of six outfall sensors for a combined 41 days. Severity: Medium. The sensor failures were caused by water ingress into the telemetry enclosures. Because no daily health check exists, the first failure was not noticed for nineteen days. Manual flow estimates were used for the affected periods.

Finding 3: Spill response kits at four of nine inspection points were missing absorbent booms. Severity: Medium. The booms had been used during a hydraulic oil spill in December and were not replaced. The monthly kit inspection required by the pollution prevention plan had not been performed since November.

Finding 4: Coastal Environmental Testing delivered laboratory results within the 30-day reporting window of the permit in every month. Severity: Low. Results were received on average 14 days after sampling. The assessment notes this as an effective control.

Recommendations

Recommendation 1: Install a treatment vault on the Berth 14 Container Terminal outfall that drains the container stacking area. Priority: High. The vault should use media filtration designed to remove dissolved zinc, and design should begin immediately so that the vault is operating before the next wet season.

Recommendation 2: Add a daily telemetry health check to the Harborwatch Monitoring System and replace both failed outfall sensors. Priority: Medium. The replacement enclosures should be rated for submersion and the health check should raise an alert to the environmental coordinator when any sensor has not reported for four hours.

Recommendation 3: Restock all spill response kits under the Stormwater Management Program and inspect them monthly. Priority: Low. The inspection should be recorded in the terminal's maintenance system so that missed inspections are visible.

Financial Impact

The treatment vault capital cost for the container stacking area outfall is estimated at \$1,875,000 including design and permitting. The authority intends to fund the vault from the capital reserve and to seek a state clean water grant for part of the cost.

The Stormwater Management Program annual budget is USD 640,000, which covers sampling, laboratory fees, telemetry maintenance and training. The sensor replacements and additional inspections recommended in this report can be absorbed within the existing budget.

Appendix

Appendix A: Corrective action plan. Recommendation 1 is owned by the Chief Engineer with design complete by 31 July 2026 and construction complete by 30 November 2026. Recommendation 2 is owned by the Environmental Coordinator with a target date of 31 May 2026. Recommendation 3 is owned by the Terminal Manager with a target date of 30 April 2026.

Appendix B: Sample results summary. Zinc concentrations at the container stacking area outfall ranged from 0.08 to 0.31 milligrams per litre during the permit year against a limit of 0.12 milligrams per litre. All other outfalls remained below the limit in every month.

Appendix C: Distribution. This report is distributed to the Port Commission, the Executive Director, the Director of Terminal Operations and the state environmental agency.